2018 Q1

Solution	Marks	Remarks
1. (a) Amount of energy required $E = mc\Delta T$ $= 6 \times 4200 \times (50 - 15)$ $= 882000$ J (or 882 kJ)	1M	
Power $P = \dfrac{E}{t} = \dfrac{882000}{60}$	1M	
$= 14700$ W (or 14.7 kW)	1A	
	3	
(b) Let m kg per minute be the water flow rate $mc\Delta T = Pt$ $m(4200)(40 - 15) = 14700 \times 60$	1M	Or $\dfrac{m}{t} \propto \dfrac{1}{\Delta T}$
$m = 8.4$ (kg min^{-1} or kg)	1A	$\therefore \dfrac{m}{6} = \dfrac{50-15}{40-15}$
	2	
2. (a) $n = \dfrac{pV}{RT} \propto pV$ (for constant T) $\dfrac{n_A}{n_B} = \dfrac{(p)(3V)}{(2p)(2V)}$	1M	
$n_A = 0.75 \times 0.8$ mol $= 0.6$ (mol)	1A	

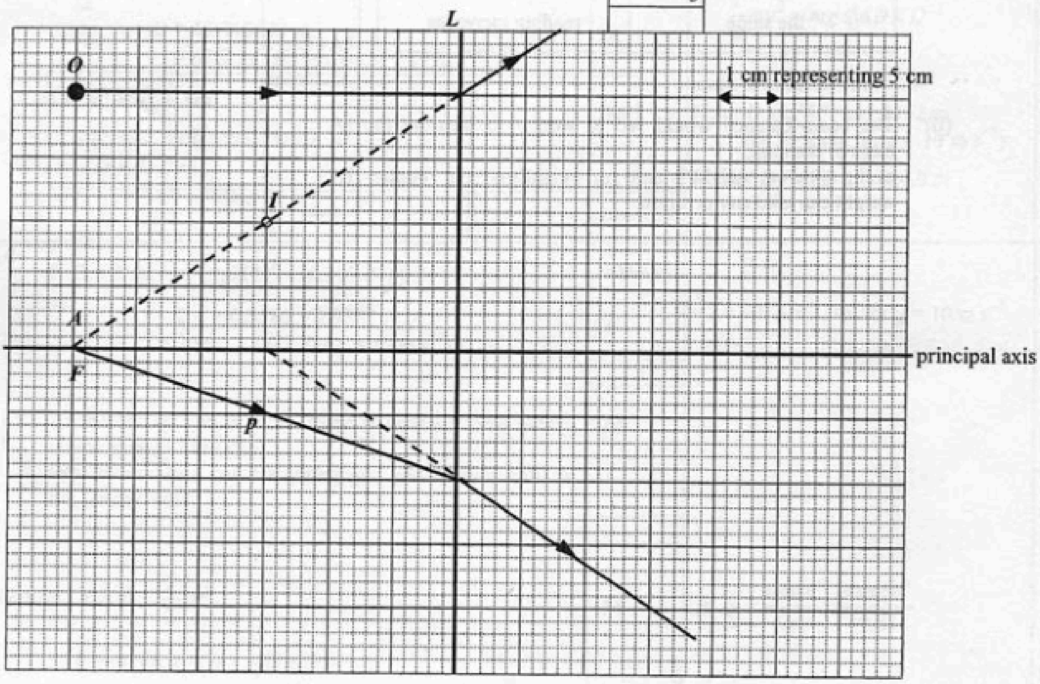
2018 Q2

Solution	Marks	Remarks
Or $(2p)(2V) = 0.8RT$ $pV = 0.2RT$ $p(3V) = nRT$	1M	
$n = 3 \times 0.2 = 0.6$ (mol)	1A	
	2	
(b) (i) $n = n_A + n_B$ $p'(3V + 2V) = p(3V) + (2p)(2V)$	1M	
$p' = 1.4p$	1A	
Or $p'(3V + 2V) = (0.6 + 0.8)RT$ $p'(5V) = 1.4RT$	1M	
$p' = 1.4p$	1A	
	2	
(ii) When the tap is open, number of gas molecules in vessel A increases due to the (net) flow of molecules from B to A,	1A	
the frequency of collision of gas molecules with the vessel's wall increases, pressure increases.	1A	
	2	

	Solution	Marks	Remarks
3. (a)	If the maximum load is exceeded, the braking distance will increase if the friction provided remains the same.	1A	
	Vehicles would not be able to stop in time in case of emergency (thus dangerous).	1A	
	Or A larger friction is required in braking the vehicle within the same distance,	1A	
	accident may occur if the brakes cannot provide such frictional forces.	1A	
		2	
(b) (i)	If the brakes are applied continuously, thermal energy generated will heat up the brake pads / brakes to too high a temperature that the brakes may fail.	1A	
		1	
(ii)	Let D be the distance travelled along the ramp. Kinetic energy of vehicle becomes its gravitational potential energy: $\frac{1}{2}m(25)^2 = m(9.81)(D\sin 30^\circ)$	1M	Or work done against vehicle's weight component along the ramp by its kinetic energy: $\frac{1}{2}mv^2 = mg\sin\theta \times D$
	$D = 63.710499$ m ≈ 63.7 m (62.5 m for $g = 10$ m s^{-2})	1A	mass of vehicle = m
	Or $v^2 = u^2 + 2as$	1M	Note: $D\sin 30^\circ = 31.8552$ m

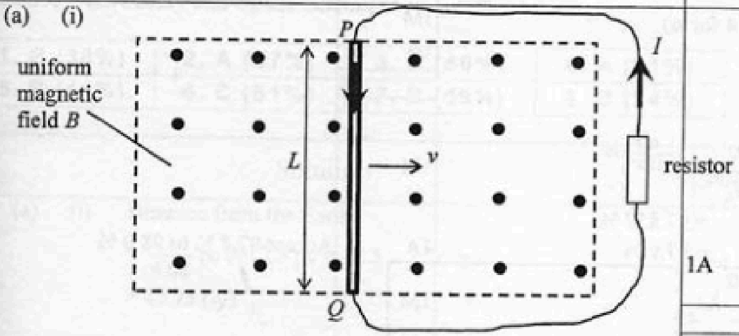
	Solution	Marks	Remarks
	$0^2 = 25^2 + 2(-9.81\sin 30^\circ)D$		(or 31.25 m for $g = 10$ m s^{-2})
	$D \approx 63.7$ m	1A	Accept $D = 62.0$ m to 64.0 m
		2	
4. (a) (i)	K.E. + P.E. $= \frac{1}{2}(0.3)(4)^2 + (0.3)(9.81)(0.2)$	1M+1M	
	$= 2.4 + 0.5586 = 2.9886$ J ≈ 2.99 J (3.0 J for $g = 10$ m s^{-2})	1A	$= 2.4 + 0.6 = 3.0$ J for $g = 10$ m s^{-2}
		3	
(ii)	As the spring gun is fixed, there is external force acting on the system / the gun,	1M	
	total momentum (of the spring gun and cannon ball) is not conserved.	1A	
		2	
(b)	Vertical : $s = ut + \frac{1}{2}at^2$ $0 = (4\sin 50^\circ)t_f - \frac{1}{2}(9.81)t_f^2$	1M	Or $\frac{t_f}{2} = \frac{u\sin\theta}{g} = \frac{4\sin 50^\circ}{9.81}$
	$t_f = 0.624705$ s (0.612836 s for $g = 10$ m s^{-2}) ≈ 0.625 s (0.613 s for $g = 10$ m s^{-2})	1A	Accept $t_f = 0.61$ s to 0.63 s
	Horizontal : $R = 4\cos 50^\circ \times t_f = 4\cos 50^\circ \times 0.625$	1M	Or $R = u\cos\theta \times \frac{2(u\sin\theta)}{g}$
	$= 1.606210$ m ≈ 1.61 m (1.57 m for $g = 10$ m s^{-2})	1A	Accept $R = 1.56$ m to 1.62 m
		4	
(c)	t_f increases	1A	
	since the initial vertical velocity / component is greater.	1A	
		2	

<table>
<tr><th>Solution</th><th>Marks</th><th>Remarks</th></tr>
<tr><td>5. (a) (i) m (5.0 cm) = 50 g (10.0 cm)
m = 100 g or 0.1 kg</td><td>1M
1A
2</td><td></td></tr>
<tr><td>(ii) Counter-weight position : 10.0 cm ± 0.1 cm
Percentage error = $100\% \times (\frac{0.1}{10.0}) = 1\%$
$\therefore m$ = 101 g to 99 g
i.e. maximum error = ± 1 g</td><td>1M
1A
2</td><td></td></tr>
<tr><td>(b) Spring balance reading = mg = (0.1 kg) (9.81 N kg^{-1})
= 0.981 N (1.0 N for g = 10 N kg^{-1})
≈ 1.0 N</td><td>1M/1A
1</td><td></td></tr>
<tr><td>(c) (i)
<table>
<tr><th>counter-weight position on beam balance</th><th>spring balance reading</th></tr>
<tr><td>the same</td><td>reading increases</td></tr>
</table></td><td>1A+1A
2</td><td></td></tr>
<tr><td>(ii) The beam balance would fail to work / to measure the mass of the load,
as the apparent weight is zero (or weightless), the counter-weight can take any position.</td><td>1A
1A
2</td><td></td></tr>
</table>

		Marks	Remarks
6. (a)	Place the (cylindrical) lens on the piece of paper and trace its outline, use ray tracing method e.g. - direct and trace a light ray towards the lens. - direct and trace a light ray along / parallel to the principal axis	1A	Or Using lens formula: - use ray tracing method: direct and trace a light ray (not parallel to the principal axis) towards the lens
	Direct another light ray parallel to the first one / principal axis (by shifting the ray box) and trace the path(s) of the (emerging) ray(s) on the paper.	1A	- extend (the path of) the emerging ray backward and locate the point of intersection (with the principal axis).
	Extend (the path of) the emerging rays (backward) and locate the point of intersection (on the focal plane containing F).	1A	- measure the corresponding object distance u and image distance v (on the principal axis)
	Measure the distance from the point of intersection (or F) to the centre of the lens, which gives the focal length of the lens.	1A	- sub. u and v into lens formula to find f
	Source of error: Scale uncertainty/accuracy/precision of the plastic ruler (read to nearest mm). OR Unable to mark the path correctly because of the thickness of the beam of light from the ray box OR The ray(s) is/are not parallel (to the principal axis) OR Any reasonable answer	1A	Accept: Using diagram(s) to supplement / simplify the description.
		5	
(b)	 		
	(i) L is diverging/concave. - only a diverging lens can produce a (virtual / erect / diminished) image between the object and the lens	1A 1A	
		2	
	(ii) Focal length = 30 cm Correct ray to find F	1A 1A	Accept (30 ± 1) cm
		2	
	(iii) Correct ray p	1A	
		1	

	Solution	Marks	Remarks
7. (a) (i)	Increase the separation between the double slit and the screen, D.	1A	
		1	
(ii)	The separation of the bright dots on the screen becomes larger, thus the percentage error in its measurement is smaller.	1A	
		1	
(iii)	The angular position of the 2nd order bright dot $\theta = \tan^{-1}\left(\frac{1.56/2}{1.40}\right) = 29.124053°$	1M	
	Grating spacing $d = \frac{10^{-3}}{400} = 2.5 \times 10^{-6}$ m	1M	
	Applying $d \sin\theta = n\lambda$, Wavelength $\lambda = \frac{2.5\times10^{-6}\times\sin 29.12°}{2}$ $= 6.08378 \times 10^{-7}$ m $\approx 6.08\times10^{-7}$ m (= 608 nm)	1A	Accept $\lambda = (6.06 \text{ to } 6.10)\times10^{-7}$ m
		3	
(b) (i)	The equation can only be applied for - $\lambda \ll a$ (i.e. wavelength << separation of the two sources), OR λ is much smaller than a - $a \ll D$ (i.e. separation of the two sources << separation of the sources and detector), OR a is much smaller than D Any ONE	1A	Note : the order of magnitude of the wavelength of sound is about 10^{-1} m
	Or Using the fringe separation equation to find the wavelength of sound is not accurate for - λ is comparable to/ NOT much smaller than a - a is NOT much smaller than D Any ONE	1A	
		1	
(ii)	For the 1st order maximum, wavelength λ = path difference $PB - PA$	1M	
	$= \sqrt{(1+0.5)^2+2^2} - \sqrt{(1-0.5)^2+2^2}$ $= 2.5 - 2.06155281 = 0.43844719 \text{ m} \approx 0.438$ m	1M	
	speed of sound: $v = f\lambda = 750 \times 0.4384$ $= 328.835$ m s^{-1} ≈ 329 m s^{-1}	1A	Accept $v = 328$ m s^{-1} to 330 m s^{-1}
		3	

Solution	Marks	Remarks
8. (a) (i) To terminals X	1A	
	1	
(ii) $P = IV$ $800\text{ W} = I(220\text{ V})$ $I = 3.636364\text{ A}$ $\approx 3.64\text{ A}$	1M 1A	
	2	
(iii) $800 = \dfrac{V^2}{R} + \dfrac{V^2}{4R} = \dfrac{5V^2}{4R}$	1M	
$P_{\text{keep warm}} = \dfrac{V^2}{4R}$	1M	
$= 800(\frac{1}{5}) = 160\text{ W}$	1A	
Alternative (I) $800 = \dfrac{V^2}{R} + \dfrac{V^2}{4R} = \dfrac{5V^2}{4R}$ $R = 75.625\ \Omega$	1M	
$P_{\text{keep warm}} = \dfrac{V^2}{4R}$	1M	
$= \dfrac{220^2}{4(75.625)}$ $= 160\text{ W}$	1A	
Alternative (II) Power $\propto \dfrac{1}{\text{resistance}}$	1M	
$\dfrac{P_{\text{keep warm}}}{P_{\text{heating}}} = \dfrac{R_{\text{eq}}}{4R}$ $P_{\text{keep warm}} = P_{\text{heating}}\left(\left(\dfrac{1}{R}+\dfrac{1}{4R}\right)^{-1} / 4R\right)$	1M	
$= 160\text{ W}$	1A	
	3	
(b) (i) Electrical energy (consumed)	1A	
	1	
(ii) (1) only the fuse blows (2) only the RCCB cuts off the supply	1A 1A	
	2	

	Solution	Marks	Remarks
9. (a) (i)		1A	
		1	
(ii)	By Lenz's law, a magnetic force F_B acts on the rod such that it opposes the rod's motion,	1A	Accept : Work done by an external force is needed to transfer mechanical energy into electrical energy.
	an external force F is needed to balance F_B so as to maintain uniform motion (or constant v)	1A	
	$\therefore F = F_B = ILB$ (in magnitude)	1A	
	Or $F \cdot v = I\xi$, $\therefore F = \dfrac{I\xi}{v}$	1A	
		3	
(iii)	mechanical power input $= Fv$ $= (ILB)\,v$	1M	
	power input = power output (electrical) $ILB\,v = I\,\xi$ $\xi = BLv$	1M	
		2	
(b) (i)	Horizontal (component) is perpendicular to the mast.	1A	
	Or Vertical (component) is parallel to the mast.	1A	
		1	
(ii)	$\xi = (B\cos30°)\,L\,v$ $= (50\times10^{-6}\cos30°)\,(20)\,(6)$	1M	by (a)(iii) using the horizontal component of B
	$= 5.196152\times10^{-3}$ V ≈ 5.20 mV	1A	
	more electrons at end X	1A	
		3	
(iii)	No current.	1A	
	Both the cable and the mast cut the field lines in the same way, the e.m.f.'s produced are equal and thus oppose each other.	1A	
	Or No change in magnetic flux through the loop of the circuit.	1A	
		2	

10. (a)	(i)	$226 - 206 = 20$ (multiple of 4 for α)	1M		
		$\therefore {}^{206}_{82}\text{Pb}$ is the end product	1A		
				2	
	(ii)	% undecayed Ra-226 left $\dfrac{N}{N_0} = e^{-\frac{\ell n 2}{1600}\times 50}$	1M		
		$= 97.857\ \%$			
		$\approx 97.9\ \%$	1A		Accept 97.8 % to 98.0 %
		Or % undecayed Ra-226 left			
		$= \left(\dfrac{1}{2}\right)^{\frac{50}{1600}}$	1M		
		$\approx 97.9\ \%$	1A		
				2	
(b)	(i)	$\because$ random process	1A		
				1	
	(ii)	Some of the daughter products of Ra-226 are also radioactive and may emit β particles.	1A		
				1	
	(iii)	Reason: weaker ionizing power of β and γ radiations	1A		
		- raise the source to a distance greater than the range of α (a few cm) will cease to have sparks. - insert a paper between the source and the gauze, sparks will cease to produce. — Any ONE	1A		
				2	